

Babak Raeisi

babak.raeisi.brs@gmail.com | (416) 986-3643 | Vaughan, Ontario, Canada | LinkedIn | GitHub

Professional Summary

Software Developer with experience building C#/.NET applications, ASP.NET Core APIs, database-driven systems, and interactive applications. Experienced with PostgreSQL, SQL Server, MongoDB, Redis, Kafka, Docker, AWS, React, and automated CI/CD workflows. Strong interest in backend engineering, distributed systems, gameplay systems, and real-time interactive software, with professional experience developing and maintaining C# applications.

Technical Skills

- **Languages:** C#, JavaScript, TypeScript, SQL
- **Backend:** .NET, ASP.NET Core, Web API, REST APIs, Entity Framework Core, LINQ, Dapper
- **Frontend:** React, TypeScript, JavaScript, HTML, CSS
- **Game & Interactive Development:** Unity, Unreal Engine, Game AI, A* Pathfinding
- **Databases & Caching:** PostgreSQL, SQL Server, MongoDB, Redis
- **Distributed Systems:** Microservices, Apache Kafka, Background Workers, Event-Driven Architecture
- **Cloud & DevOps:** AWS, Docker, Docker Compose, GitHub Actions, CI/CD
- **Architecture & Practices:** OOP, SOLID, Dependency Injection, Repository Pattern, Design Patterns
- **Security & Testing:** JWT Authentication, Authorization, xUnit, NUnit
- **Tools:** Git, Visual Studio, VS Code, Postman, JIRA

Professional Experience

Faza Zaman Hormozgan

Software Developer

Dec 2019 – May 2022

Bandar Abbas, Iran

- Developed and maintained C# applications and interactive Unity-based software integrated with backend services and SQL Server.
- Built application features involving REST API communication, JSON data processing, and database-driven workflows.
- Developed interactive 3D and AR functionality using Unity and C#, including user interfaces and application logic.
- Diagnosed defects, refactored existing components, and improved maintainability of production application code.
- Collaborated within Agile development workflows using JIRA for issue tracking and sprint management.

Pishgaman Omid Darya Shipping

IT Assistant & Software Developer

Aug 2018 – Dec 2019

Bandar Abbas, Iran

- Developed a C# WPF desktop application to automate crew-list generation, invoicing, and operational data workflows.
- Designed and maintained SQL Server databases used for data entry, reporting, and shipping operations.
- Troubleshoot application and database issues and extended software functionality as business requirements changed.
- Built maintainable C# components for business-critical desktop workflows.

Selected Projects

MicroBooker – Distributed Reservation System

2026

ASP.NET Core, C#, React, PostgreSQL, MongoDB, Redis, Kafka, Docker, AWS

- Built a distributed restaurant reservation platform with React and separate ASP.NET Core authentication

and reservation services.

- Implemented Redis distributed locking and a MongoDB unique compound index to protect reservation slots during concurrent requests.
- Integrated Kafka with a .NET background worker for asynchronous reservation processing, idempotent persistence, and Redis caching.
- Implemented JWT authentication with server-side user identity extraction for protected reservation operations.
- Containerized services with Docker and deployed the application using AWS EC2, ECR, CloudFront, Route 53, and GitHub Actions.

Pandora – Cognitive Training Application

2025 – 2026

Unity, C#, ASP.NET Core, Entity Framework Core, PostgreSQL, Docker

- Built a Unity application containing three cognitive training mini-games with progressive difficulty across multi-day sessions.
- Designed structured trial and session tracking for gameplay results, completion times, errors, and difficulty progression.
- Developed an ASP.NET Core analytics API with Entity Framework Core and PostgreSQL for storing and querying player sessions and trials.
- Implemented Unity-to-API JSON uploads with request validation, payload normalization, timeout handling, and network failure handling.
- Containerized the analytics backend with Docker for local and cloud-oriented deployment.

HopHovac – Unity Arena Game

2025

Unity, C#, Game AI, A Pathfinding, Grid Systems*

- Built a grid-based 3D arena game with dynamic territory ownership, player movement, pickups, scoring, and AI-controlled opponents.
- Implemented A* pathfinding with dynamic route recalculation when movement paths became blocked.
- Designed AI behavior around Explore, Collectable, and Compete strategies with runtime strategy evaluation and switching.
- Created a shared grid and node system for occupancy, ownership, world-to-grid conversion, movement validation, and gameplay state.
- Implemented object pooling for collectibles and reusable gameplay systems for player and AI movement.

Certifications

- AWS Certified Cloud Practitioner – 2025
- Microsoft Backend Development – 2025
- IBM Agile Development & Scrum – 2022

Education

- **Master Diploma in Video Game Art**, Universal Art School, Valencia, Spain, 2022–2024
- **B.Sc. Computer Engineering – Software**, Islamic Azad University, Iran, 2014–2019 WES Evaluated